

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Converting ly to m i.e. $200 \times 9.46 \times 10^{15} = 1.892 \times 10^{18}$ [m] (1) Calculating θ in radian i.e. $\frac{1.5 \times 10^{11}}{1.892 \times 10^{18}} = 7.93 \times 10^{-8}$ (1) (can be done by using tan or trig) $\theta = 1.22 \frac{\lambda}{d}$ used (whatever the subject) i.e. 7.6 m for visible light or 7.625×10^{-8} seen (1) Final calculation leading to suitable conclusion (1) Award full marks for a calculation of 1.45×10^{11} seen and compared with 1 AU	1	1 1		4	3	
	(b)	(i)	Substitution into period equation e.g. even $50 = 2\pi \sqrt{\frac{d^3}{G \times 3 \times 2 \times 10^{30}}}$ (1) Final rearrangement and correct substitution seen or 2.93×10^{12} [m] (1)	1	1		2	1	
		(ii)	Substitution into C of M equation (1) can be implied $\frac{1}{3} \times 2.93 \times 10^{12} = 0.977 \times 10^{12}$ or equivalent seen (1)	1	1		2	1	
		(iii)	Use of Stefan's law (1) Use of area proportional to R^2 (1) $\frac{R_A^2 T_A^4}{R_B^2 T_B^4} = \frac{R_A^2 1.7^4}{R_B^2 4.3^4} = \frac{25}{1/18}$ (1) Answer = $\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}} = 136$ or 140 (1)	1	1 1 1		4	3	